Step 1: battery to ammeter

Connect the battery positive terminal to the ammeter positive (red) terminal. Red is positive, black is negative. Keep every connection tight. MARKER-WIRE-P1.

Step 2: ammeter to resistor to rheostat

Join the ammeter negative terminal to one end of the unknown resistor, then through the rheostat back to the battery negative terminal. The rheostat varies the current. MARKER-WIRE-P2.

Step 3: voltmeter in parallel

Connect the voltmeter across the resistor only: red (positive) where current enters, black (negative) where it leaves. A voltmeter is always in parallel. MARKER-WIRE-P3.

Connection table:

from	**to**	**wire**
battery +	ammeter + (red)	tight
ammeter -	resistor end A	tight
resistor end B	rheostat	tight
rheostat	battery -	tight
voltmeter red	resistor end A	parallel
voltmeter black	resistor end B	parallel

Step 4: read and record

Trace the loop: battery+ to ammeter+ to resistor to rheostat to battery-. Vary the rheostat, record each voltmeter/ammeter pair, and check V/I stays constant. MARKER-WIRE-P4.

Appendix: observation log (20 runs, 12V rail)

Rheostat stepped down run by run; V/I stays near 250 ohm throughout.

run	rheostat	voltage_V	current_mA
1	high	12.0	48.0
2	mid	11.5	46.0
3	low	11.0	44.0
4	high	10.5	42.0
5	mid	10.0	40.0
6	low	12.0	48.0
7	high	11.5	46.0
8	mid	11.0	44.0
9	low	10.5	42.0
10	high	10.0	40.0
11	mid	12.0	48.0
12	low	11.5	46.0
13	high	11.0	44.0
14	mid	10.5	42.0
15	low	10.0	40.0
16	high	12.0	48.0
17	mid	11.5	46.0
18	low	11.0	44.0
19	high	10.5	42.0
20	mid	10.0	40.0